

Multimodal Federated Tea Disease Diagnosis

Ayush Debnath

Ruelia Saha

Sudip Misra

Damodhara R. Mailapalli

Abstract—Tea pest and disease diagnosis benefits from lesion imagery and field notes, while federated adaptation can keep both records on-site. Small agricultural corpora, however, make multimodal and privacy claims easy to overstate. Our proposed model is a leakage-controlled five-class vision-language system with source-grouped splitting, exact image-box/text linkage, missing-modality tests, reliability-aware fusion, and measured FedAvg adaptation. The corpus contains 371 crops from 200 photographs, split into 222/74/75 train/validation/test crops from 122/40/38 disjoint sources; all 371 text records link to the exact box, and a training-fitted filter masks 178 target-shortcut tokens. On the identical 75-crop internal support under deterministic 50%-retention notes, frozen DistilBERT text-only, frozen ViT-tiny image-only, the raw ResNet–Transformer VLM, a new frozen ViT-tiny–DistilBERT VLM, and the proposed router obtain 65.33%, 66.67%, 77.33%, 82.67%, and 81.33% accuracy. Architecture selection is validation-only: the candidate’s 82.43% validation accuracy remains below the incumbent’s 89.19%, so ResNet-50 and the compact text Transformer are retained before candidate test inference. The candidate’s one-crop internal-test edge is non-significant (exact McNemar $p = 1.0$), although its macro F1 is higher (0.8192 versus 0.6679). A separate same-checkpoint diagnostic reaches 92.00% with paired input but 33.33% after class-mismatching the text, demonstrating alignment sensitivity; its compact Transformer and ResNet paths are not relabelled as independent PLM and ViT baselines. Validation-only adaptation retains 98.1–100.0% of the 0.9099 centralized macro F1 over 2–8 clients. A separate 1,000-pair matched-setting stress test finds FedAvg macro F1 of 0.715/0.809 at $\alpha = 0.1/1$, warm-start 0.852 versus cold-start 0.809, and 59.5–60.5 MiB upload per client-round. Those pairs are label-matched resamples rather than co-observations, clients are simulated, and multi-site validation remains necessary before privacy or deployment claims.

Index Terms—Federated Learning, Vision-Language Models, Tea Pest and Disease Classification, Multimodal Learning, Evaluation Protocol, Data Leakage, Precision Agriculture

I. INTRODUCTION

TEA (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five visually confusable stress classes — leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug — reduce yield and leaf quality across major growing regions [1], [2]. Recent diagnostic systems improve field deployment through lightweight CNNs, attention modules, and edge-oriented pipelines [1]–[3], [5], [6], [8], but nearly all begin and end with the photograph. That is not how tea diagnosis is recorded on the ground, where staff note lesion shape, flush condition, weather history, and pest pressure alongside the image. Meanwhile, sending every high-resolution image and field note to a central repository is unattractive for estates with limited connectivity and proprietary records. Federated Learning [15] addresses the second problem, but existing FL work on tea pathology has not integrated the visual and textual evidence used in real inspections.

A second, methodological motivation became dominant during this study. One source photograph may contain several annotated regions, so crop-level random splits place near-identical leaf context on both sides of the boundary. Text can leak the target even more directly when disease names or class-pure terms remain in descriptions. We therefore group

TABLE I: Comparison with Related Work.

Work	Image	Text	FL	VLM	Leak. audit
Yang et al. [1], Hu et al. [2], Deng et al. [3]	✓	×	×	×	×
Joseph et al. [5], Maurya et al. [6]	✓	×	×	×	×
Tahir et al. [10], Devaraj et al. [11]	✓	×	×	×	×
Ding et al. [12], Caminha et al. [13]	✓	×	✓	×	×
Wang et al. [27]	✓	✓	×	✓	×
Li et al. [28]	✓	✓	✓	✓	×
This work	✓	✓	✓	✓	✓

by source image, link text at the exact box level, and learn the blocked vocabulary from training annotations only. This changes the scientific question from “can a model memorise a disease description?” to “does each modality contribute on unseen source leaves?”

Contributions. We provide (i) exact crop–note linkage with source-disjoint splits and train-only leakage control; (ii) common-support unimodal, paired, mismatched, and model-family comparisons; (iii) explicit visual and sparse-note routing; and (iv) three-seed client scaling plus a matched reviewer suite covering partitioning, anti-collapse components, warm start, fusion, communication, and four FL aggregators against local-only.

II. RELATED WORK

Tea diagnosis has progressed from backbone comparisons to lightweight, attention-guided and IoT-assisted image systems [1]–[3], [5]–[9], [38]–[41], [45], [46], [48], [49]. Agricultural FL extends FedAvg [15] with explainability, hierarchies, blockchain, cloud, and edge coordination [10]–[14], while deployment raises trust and continual-update concerns [42], [47]. VLMs introduce contrastive, query-former, gated-attention, and joint objectives [22]–[25]; recent crop systems combine these ideas with multimodal diagnosis or federated transfer [27], [28], [43], [44]. Table I highlights the remaining gap: exactly paired image–text FL with a source-disjoint leakage audit.

III. SYSTEM MODEL

Each estate keeps images and notes local; only model updates cross the boundary. The controlled ablations use one selected checkpoint, while FL adapts its trainable multimodal states with the visual backbone frozen. Text uses a stable 30 522-id hash vocabulary (length 128); normalized 224×224 images yield 49 spatial tokens for bidirectional attention.

A. Dataset and Leakage Audit

The audited corpus contains 200 field photographs collected from the IIT Kharagpur tea garden and annotated with YOLO oriented bounding boxes. One crop is extracted per valid box with 10% context padding, giving 371 examples across *LEAF_BLIGHT* (129), *LEAF_HOPPERS* (9), *LEAF_RUST* (67), *LOOPER_CATERPILLARS* (102), and *MOSQUITO_BUG* (64). Classes are the raw YOLO identifiers under the supplied annotation schema.

Seed 42 splits source filenames, never crops: overlap is zero and the test holds 75 crops from 38 unseen photographs. All 371 records link by exact (photograph, box index); the nine-example leaf-hopper class remains the bottleneck. Notes

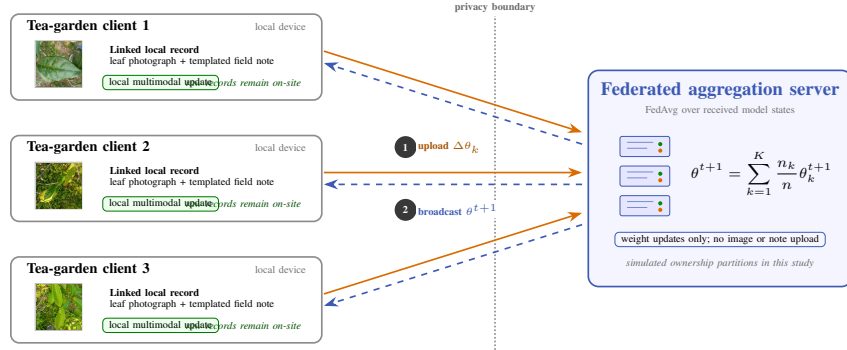


Fig. 1: Federated deployment workflow using representative corpus photographs. Each simulated tea-garden client keeps linked photographs and field notes on-site; only model updates $\Delta\theta_k$ cross the explicit privacy boundary, and the FedAvg server returns θ^{t+1} .

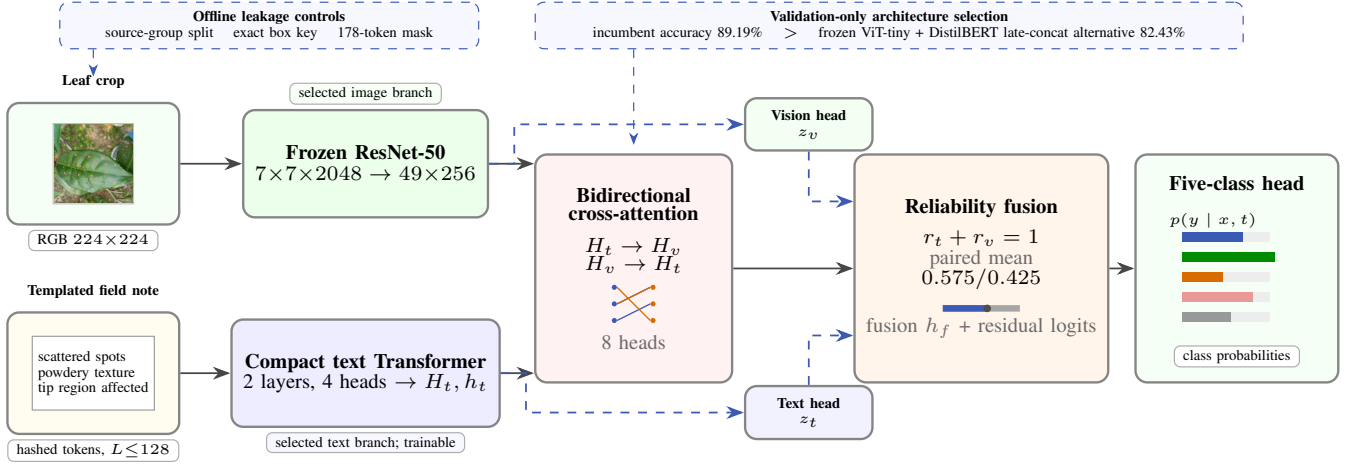


Fig. 2: Active inference architecture with a real corpus crop and a representative templated note. The dashed blue boxes separate offline leakage controls and validation-only system selection from runtime inference. The selected ResNet–compact-Transformer core uses bidirectional cross-attention; dashed runtime paths are explicit unimodal expert heads that re-enter the fusion logits. The pooled ViT–DistilBERT alternative is screened separately and is not drawn as token-level cross-attention.

TABLE II: Source-grouped paired-data audit

Split	Sources	Crops	Blight	Hoppers	Rust	Looper/Mosq.
Train	122	222	77	5	40	61/39
Validation	40	74	26	2	14	20/12
Test	38	75	26	2	13	21/13
Total	200	371	129	9	67	102/64

are templated from manually written phrase pools, not independent farmer prose. A 178-token target-pure vocabulary is learned on training text only and then frozen. No external tea images are pooled; ImageNet supplies weights only.

B. Model Variants and Objective

Table III summarizes the implementation. The reliability network produces (r_t, r_v) , $r_t + r_v = 1$, and the final logits are $\mathbf{z} = \mathbf{z}_f + 2(r_t \mathbf{z}_t + r_v \mathbf{z}_v)$, with a missing modality’s expert zeroed. The residual remains active at test time.

Comparison models. Frozen DistilBERT [33] and ViT-tiny [17] use validation-selected ridge heads. Their late-fusion candidate uses train-fitted 64-component projections and one of 360 validation-screened linear heads; test embeddings are extracted only after the choice is frozen. The raw and proposed VLM rows are the cross-attention output before and after the router in Figs. 2 and 3.

For one-hot label y , training uses class-weighted, label-smoothed cross entropy with auxiliary parent losses and supervised cross-modal alignment:

$$\mathcal{L} = \mathcal{L}_f + 0.20\mathcal{L}_t + 1.25\mathcal{L}_v + 0.05\mathcal{L}_{align}. \quad (1)$$

The visual weight is deliberately larger because image-only learning is the harder path. Modality dropout samples image-only, text-only, and paired batches with probabilities 0.50, 0.10, and 0.40. Checkpoint selection uses $0.65F_{1,\text{paired}} + 0.35F_{1,\text{image}}$ on validation.

For K clients, the same local objective is aggregated with FedAvg [15],

$$\theta^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1}, \quad (2)$$

where n_k is client k ’s sample count. FL starts from the selected checkpoint, freezes ResNet-50, and averages the remaining states after one local epoch; it is adaptation, not training from scratch.

IV. PROPOSED FRAMEWORK

A. Base Cross-Modal Checkpoint

The checkpoint trains for at most 15 AdamW epochs (batch 16, peak 10^{-4}). Two cached augmentations per training crop

TABLE III: Implemented model blueprint

Block	Specification	Output / training state
Text encoder	192-d embedding; 2 Transformer layers; 4 heads	256-d pooled vector; trainable
Vision encoder	ImageNet ResNet-50; 224 ²	49×256 tokens; frozen backbone
Cross-attention	Two directional layers; 8 heads	text-to-image and image-to-text summaries
Reliability fusion	normalized (r_t, r_v) plus interaction features	5-way trainable classifier
Expert residual	linear text and vision heads; fixed $\lambda = 2$	added to the fusion logits at train and test
Complete model	32.20M parameters	8.69M trainable parameters

TABLE IV: Class-wise visual experts selected on validation

Base class	Selected expert	Test recall
Leaf blight	Deep feature	0.692
Leaf hoppers	Calibrated base	0.500
Leaf rust	Frozen ViT	0.615
Looper caterpillars	Base TTA	0.714
Mosquito bug	Frozen ViT	0.923

remain views of 222 crops, not new samples. Selection uses 74 validation crops; the strict-reloaded checkpoint is then evaluated once on 75 test crops.

B. Enhanced Image-Only Inference

The optional visual system selects per class among the direct path, six-view TTA, a deep-feature head, calibrated logits, and frozen ViT-tiny. Validation scores all 5⁵ gates by $0.6F_1 + 0.4$ accuracy and freezes $g = [2, 3, 4, 1, 4]$ before test inference.

C. Reliability-Aware Sparse-Note Rule

For sparse notes, a stable hash retains each non-special token with probability 0.50. A validation-fixed rule routes low-confidence (< 0.60), base-predicted leaf-rust cases to the sparse-text expert; otherwise it keeps late fusion. Because the routing family was refined on this internal partition, its test score is **exploratory and non-pristine**.

D. Multimodal Federated Adaptation

FL adapts 8.69M parameters with frozen ResNet-50, one local epoch, full participation, and sample weighting. We sweep $K = \{2, 3, 5, 8\}$ for eight rounds over seeds 42, 123, and 2026 on the 74-crop validation set. Clients are simulated; the ensemble and router are excluded from this FL score.

E. Advisory Module

The Classify–Retrieve–Advise module embeds a query with Sentence-BERT [35] and performs exact FAISS inner-product search [34], [36]. Its separate templated corpus collapses 3,000 annotations to 673 observations; a deduplicated split leaves 472 passages and 201 queries with no shared observation. Correctness requires a class match.

V. PERFORMANCE EVALUATION

A. Protocol

The base checkpoint was trained in PyTorch [37] for at most 15 epochs. All crops from a source photograph remain in one partition. Model selection and the five-expert visual gate use only the 74-crop validation partition; every number below uses the same 75-crop, 38-source internal test support. Accuracy is accompanied by macro F1 because the class distribution is imbalanced. Source-group bootstrap intervals (5,000 replicates) resample source photographs rather than treating multiple crops from one photograph as independent observations.

Five evidence tracks remain separate: the exact-pair same-checkpoint ablation; the validation-selected family comparison

TABLE V: Validation-only multimodal FedAvg adaptation. Values are means over three client-partition seeds.

K	Final F1	Best F1	Retained	Label TV
2	0.8924	0.9099	98.1%	0.139
3	0.9046	0.9237	99.4%	0.217
5	0.9099	0.9297	100.0%	0.270
8	0.9099	0.9099	100.0%	0.300

on the common 75-crop support; the 74-crop validation-only FedAvg sweep; earlier text, vision, and proxy-fusion screens whose differing supports permit only within-panel ranking; and the 1,000-pair matched-setting reviewer suite. The last uses label-matched resamples and tests optimization and aggregation, not co-observation.

B. Federated Client and Round Scaling

Figure 4 and Table V answer the two federated scaling questions directly. Final-round macro F1 ranges from 0.8924 to 0.9099 and is never perfect. At the predeclared $K = 3$ setting, the final mean is 0.9046, or 99.4% of the 0.9099 initialization. Round-wise means stay between 0.8924 and 0.9216; the highest transient mean occurs for $K = 5$ at round 4, but it is reported as a validation peak rather than the final model. Increasing K raises realized label-distribution total variation from 0.139 to 0.300 without a monotone final-score penalty.

This stability result is useful but narrow. Every run begins from the same centrally selected checkpoint, so it supports federated adaptation of the multimodal core, not superiority over training from scratch. Three partition seeds measure ownership-split variability, not full checkpoint-training variance, and simulated clients cannot establish estate-level privacy.

C. Matched-Setting Reviewer Ablations

Figures 5 and 6, with Table VI, close the reviewer-requested gaps without mixing supports. At $\alpha = 0.1$, the corrected splitter realizes TV 0.571 and FedAvg F1 0.715; at fixed data, $K = 10$ reaches 0.604 versus 0.887 for $K = 2$. Warm start adds 0.043 F1, gated fusion leads, and one float32 round uploads about 60 MiB per client. Every E3 arm predicts all five classes: diversity helps the balanced arm under severe skew (0.651 versus 0.561), but the full stack is never best. FedAvg leads at $\alpha = 0.1$ and FedProx at $\alpha = 1$; SCAFFOLD is unstable. The stack is therefore a training control, not a necessary condition for avoiding collapse on this support.

D. Standard Same-Checkpoint Ablation

Paired input gives 69/75 correct versus 72/75 for text, so fusion does not beat its strongest parent. Mismatching the note drops accuracy from 92.00% to 33.33%, confirming cross-modal dependence. The family screen in Fig. 8 uses a harder sparse-note condition.

E. Raising Image-Only Accuracy

The validation-selected five-expert gate raises image accuracy from 61.33% (46/75) to **72.00%** (54/75; macro F1 0.6759). This is an ensemble, not the 66.67% pure-ViT baseline; rare-class behavior remains unstable.

F. Actual Model-Family and Inter-Model Comparison

Validation selects the incumbent before test extraction: 89.19% accuracy and 0.8541 F1 versus 82.43% and 0.6892 for the pooled ViT–DistilBERT candidate. The candidate later predicts one more test crop, but the 11-versus-10 discordance

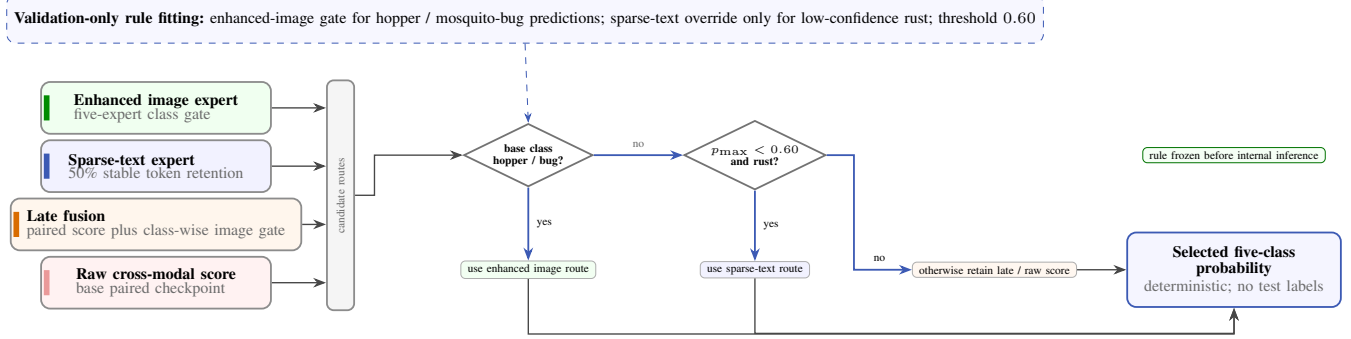


Fig. 3: Reliability-aware sparse-note routing as an explicit decision flow. Four available prediction routes feed a deterministic class/confidence rule selected on validation. Blue arrows show the frozen decisions; the router does not read test labels, and the reported internal result remains exploratory and non-pristine.

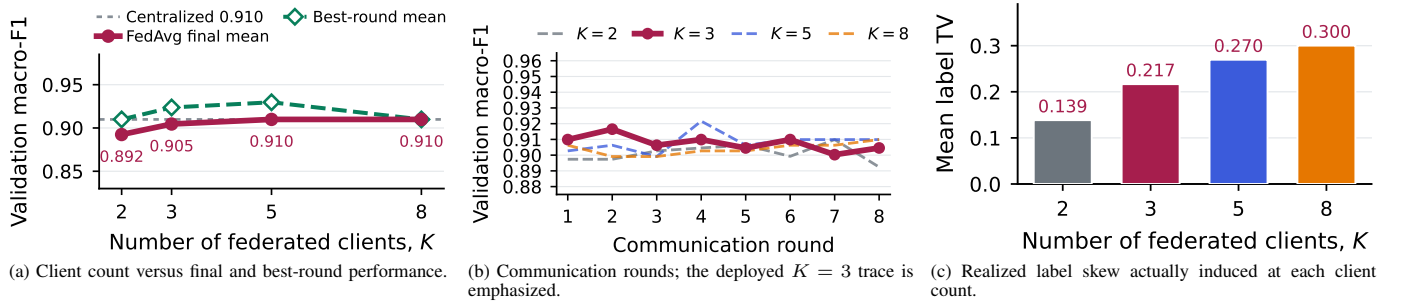


Fig. 4: Measured multimodal FedAvg adaptation on the source-disjoint tea validation split. Points are mean macro F1 over three client-partition seeds; 222 training crops, 74 validation crops, $\alpha = 1$, one local epoch, full participation, and a frozen ResNet-50 are fixed. The dotted line in (a) is the 0.9099 initialization, and (c) is the realized label total variation, so (a) is read against measured rather than assumed skew. The 75-crop test is not used, and client ownership is simulated rather than estate-derived.

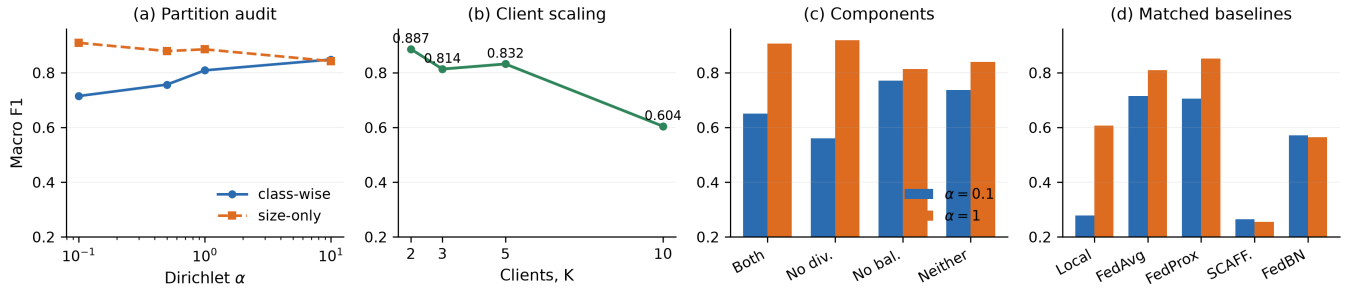


Fig. 5: Standard-tier reviewer suite. (a) The class-wise Dirichlet splitter creates the intended label skew; the legacy size-only splitter does not. (b) With fixed total data, $K = 10$ leaves much smaller client shards. (c) Anti-collapse components are ablated under actual federation. (d) Aggregators use the same backbone, partition, optimizer, and local budget. Bars in (c)–(d) compare $\alpha = 0.1$ and 1; E1/E8 average seeds 0 and 1, whereas E2/E3 use seed 0.

TABLE VI: Matched-setting reviewer results on 1,000 label-matched image–note resamples (700/150/150 split), not genuine co-observations. E8 values are two-seed means; local-only averages three clients at seed 0.

(a) Components			(b) Matched baselines			(c) Other evidence	
Configuration	$\alpha = .1$	$\alpha = 1$	Method	$\alpha = .1$	$\alpha = 1$	Evidence	Result
Both	0.651	0.906	Local only	0.279	0.607	Warm / cold start	0.852 / 0.809 F1
No diversity loss	0.561	0.919	FedAvg	0.715	0.809	Gated / attention / concat	0.890 / 0.860 / 0.809 F1
No balanced sampler	0.771	0.813	FedProx	0.706	0.852	$K = 2/3/5/10$	0.887 / 0.814 / 0.832 / 0.604 F1
Neither	0.736	0.840	SCAFFOLD	0.265	0.255	Upload C/G/A	59.5 / 60.3 / 60.5 MiB
			FedBN	0.572	0.564	Round trip C/G/A	119.1 / 120.6 / 121.1 MiB

is not significant (McNemar $p = 1.0$). Figure 8 makes the selection/test distinction visible. Its top row contains earlier screens on different supports and therefore licenses within-panel ranking only.

G. Cross-Modal Analysis and Evidence Boundary

Mean reliability is 0.575 text/0.425 image; mismatching text costs 58.67 accuracy points and 0.5949 F1. Exact cross-modal recall@1 is only 1.33%, while class-level recall@1

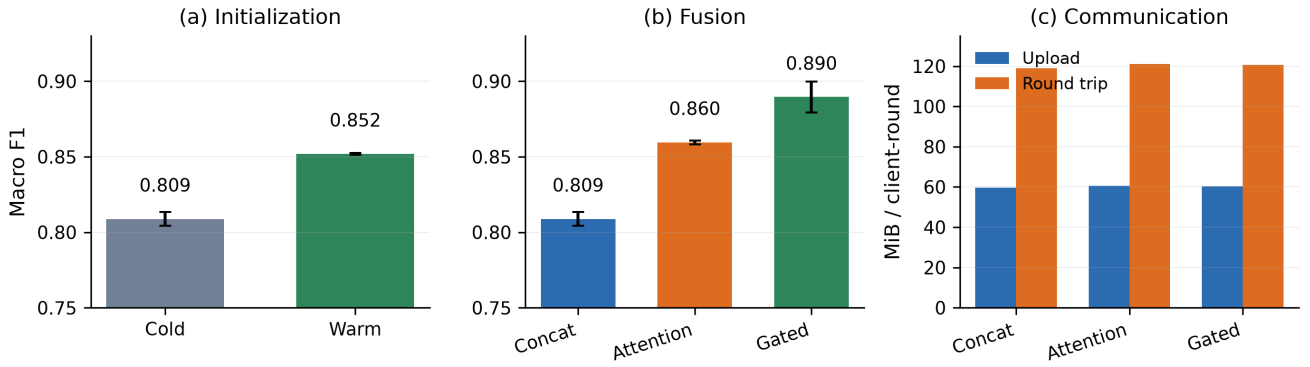


Fig. 6: Remaining matched-setting ablations (two-seed means with standard deviation bars): warm versus cold initialization, fusion strategy, and analytic float32 communication per client-round.

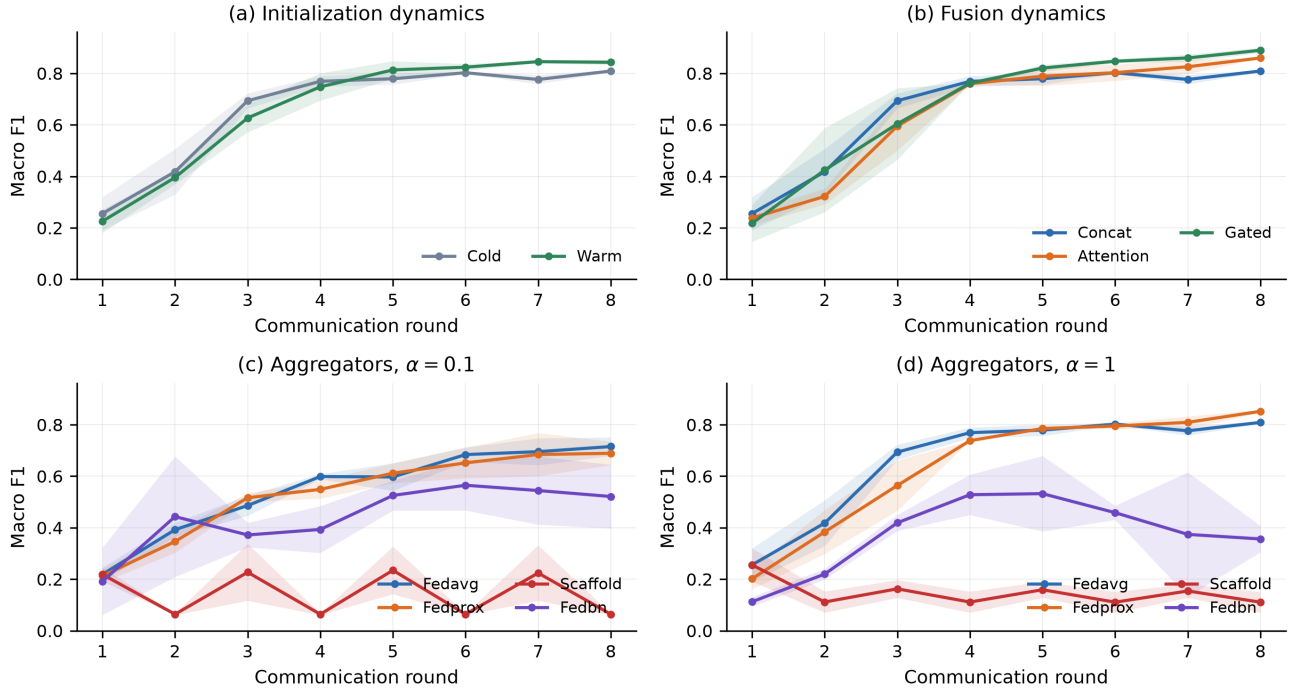


Fig. 7: Round-wise matched-setting dynamics. Lines are two-seed means and shading spans the observed seed range. Warm start separates late, gated fusion finishes highest, FedAvg/FedProx converge under both skews, and SCAFFOLD remains unstable on this support.

TABLE VII: Standard same-checkpoint ablation. These are internal core paths, not the independent PLM/ViT/VLM systems in Table VIII.

Condition	Accuracy	Correct	Macro F1	ECE
Core image path (ResNet)	0.6133	46/75	0.5431	0.1135
Core text path (compact Tr.)	0.9600	72/75	0.9049	0.1463
Base paired VLM	0.9200	69/75	0.8753	0.2648
Mismatched-pair VLM	0.3333	25/75	0.2804	0.2193

TABLE VIII: Common-support encoder selection. Architecture choice uses validation only; test metrics are reported after selection.

Actual system	Val. acc.	Test acc.	Test F1
Text PLM (DistilBERT)	0.6757	0.6533	0.6732
Image ViT (ViT-tiny)	0.7973	0.6667	0.5830
Raw ResNet+compact VLM	0.8378	0.7733	0.7155
ViT+DistilBERT VLM	0.8243	0.8267	0.8192
Proposed ResNet+compact	0.8919	0.8133	0.6679

is 0.600/0.613, indicating class organization rather than crop identity.

H. Architecture Component Ablation

Nine variants were retrained on the same split in CPU and H100 runs. Although the H100 full model reaches 94.67%, the identical CPU run gives 85.33%; the mean gap is four points and five of eight effects reverse sign. Figure 14 therefore reports instability, not a component ranking.

I. Advisory Retrieval Results

Retrieval alone returns a correct-class passage for 90.05% of queries (95% CI 85.57–94.03) against a 20.01% chance rate, with precision@5 93.23%, MRR 0.9469, and NDCG@5 0.9621. The classify stage raises this to 95.52% (95% CI 92.54–98.01), but that second number *is* the classifier’s accuracy: retrieval restricted to one predicted class can only return

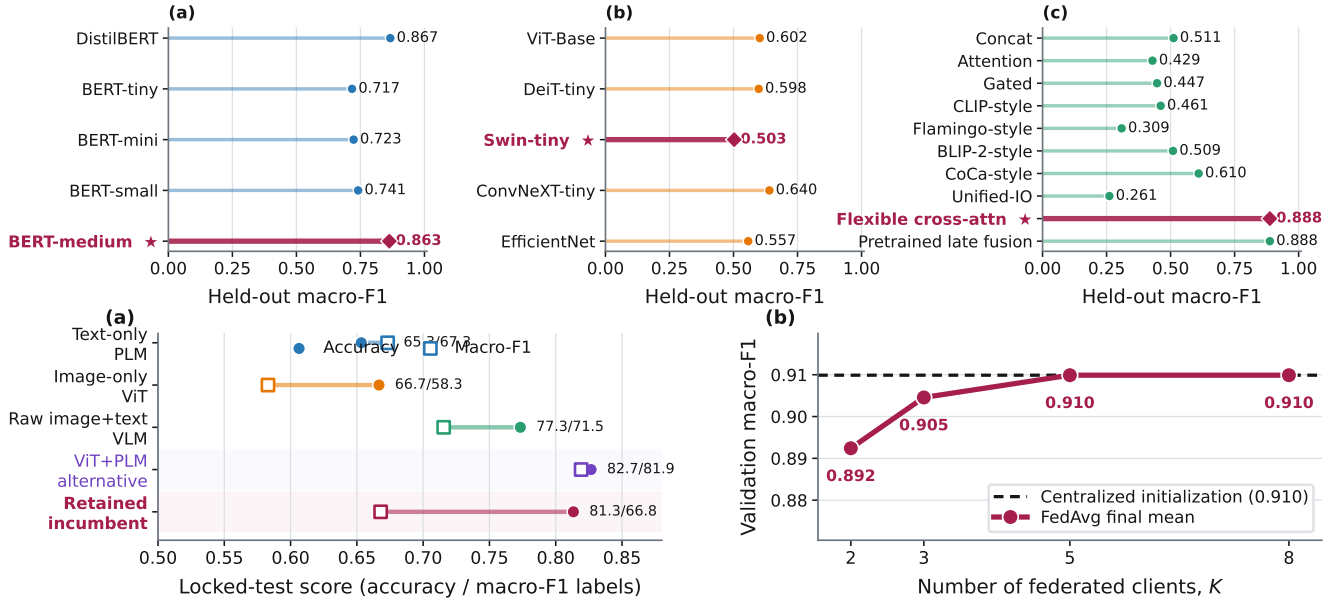


Fig. 8: Model comparisons retained as visual evidence. Top: within-family screens on their own supports (cross-panel scores are not comparable). Bottom: common 75-crop support for selected PLM, ViT, raw VLM, candidate VLM, and retained system (left), plus three-seed warm-start FedAvg scaling (right).

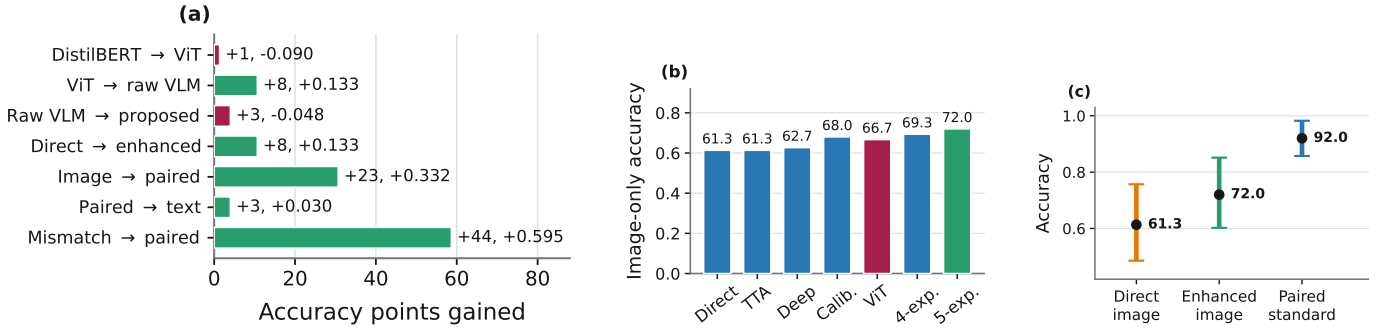


Fig. 9: Diagnostic evidence on the locked 75-crop support: (a) score gains, (b) image-expert progression, and (c) grouped bootstrap intervals. Visual gains are concentrated rather than uniform.

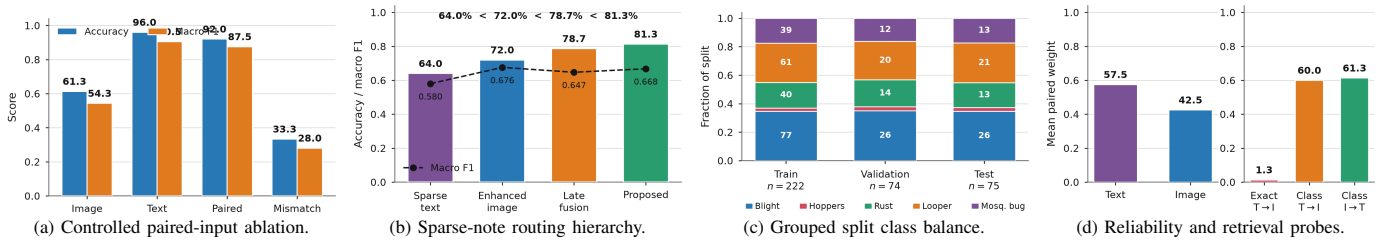


Fig. 10: Compact diagnostic panels. (a) and (c) are controlled audited data; (b) is exploratory; (d) distinguishes class-level alignment from exact-pair retrieval. Metrics with different supports are not pooled.

that class, so routing rather than ranking decides correctness. Fig. 11(b) locates the failure. On the 65.67% of queries carrying a class-unique sentence open retrieval is near-saturated at 99.24%; on the rest it falls to 72.46%, which routing lifts to 89.86%. Per class it is weakest on looper caterpillars (77.50%), whose vocabulary overlaps mosquito bug damage.

Three limits apply. The corpus is template-composed, so base and queries share sentence vocabulary even though no observation is shared; these are upper bounds relative to free-form notes. The fusion re-ranker needs a fusion model on these

five classes and is still unevaluated, as are the OBB overlay and the agronomic correctness of the advice itself. This is a retrieval result, not evidence that the recommended treatment is right.

J. Reproducibility Manifest

The updated bundle records data identity, checkpoint state, every federated round, and each realized client partition. Table IX separates mechanically verified controls from properties outside the evidence. Deterministic grouping, strict reload,

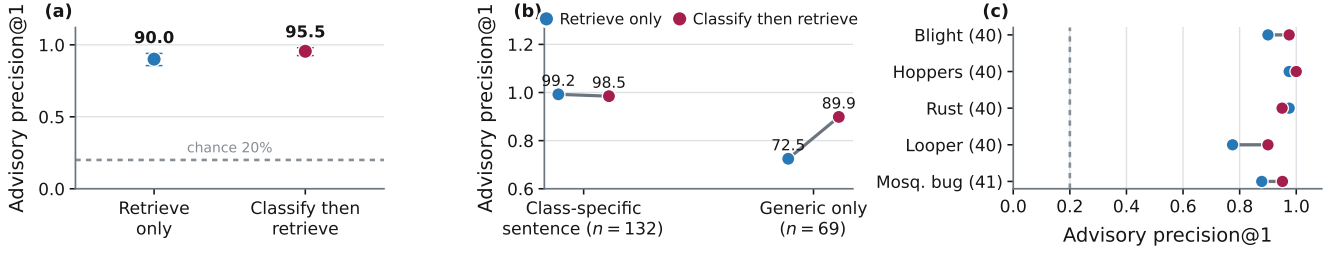


Fig. 11: Measured Classify-Retrieve-Advise quality on the templated symptom corpus: 472 passages, 201 queries, exact inner-product search. (a) Precision@1 with bootstrap intervals against chance; (b) split by whether the query holds a class-unique sentence; (c) per class. Same five classes as the diagnostic experiments, but a different corpus and support, never pooled with them.

and validation-only adaptation are reproducible; external estate transfer is not inferred from them.

TABLE IX: Reproducibility manifest for the updated run

Control	Recorded value	Status
Source split	seed 42; 122/40/38 sources	fixed
Cross-split overlap	0 source photographs	verified
Crop-text pairing	371/371 exact box keys	verified
Shortcut vocabulary	178 tokens; train fitted	fixed
Checkpoint reload	strict key match	passed
Architecture ablation	9 variants; CPU/H100 re-runs	single seed each
DistilBERT selection	90 candidates; validation only	fixed before test embed.
ViT+PLM VLM screen	360 candidates; validation accuracy	incumbent retained
Visual gate search	3,125 gates on validation	completed
Internal test support	75 crops / 38 sources	common
FedAvg adaptation	$K = 2, 3, 5, 8$; 8 rounds	completed
FL repetitions	seeds 42/123/2026	partition seeds
FL evaluation	74 validation crops	test untouched
Reviewer suite	1,000 pairs; 700/150/150 split	label-matched proxy
Matched FL baselines	4 aggregators + local-only	completed; no errors

K. Class-Level Error Accounting

The enhanced gate gains eight correct crops and the exploratory router gains 15, mainly in blight and looper; two leaf-hopper examples remain too few for a stable claim.

L. Threats to Validity

The locked test is small (38 sources, 75 crops, two leaf-hopper crops), notes are templated, and selecting five experts on 74 validation crops can overfit. The central checkpoint has one seed; FL seeds vary only client partitions. Clients are simulated, and the reviewer suite uses class-matched resamples with synthetic image fill rather than co-observations; E2/E3 use one seed. The router result is non-pristine, DistilBERT is a PLM rather than a generative LLM, and earlier fusion screens have incomparable supports. Deployment requires multi-estate data, independent notes, rare-class coverage, and repeated end-to-end training.

VI. CONCLUSION

On 371 linked crops, validation retains the ResNet-compact-Transformer router; the ViT-DistilBERT candidate's one-crop test edge is non-significant ($p = 1.0$), and CPU/H100 component effects are unstable. Warm-start FedAvg retains 98.1–100.0% of initialization over 2–8 simulated clients. In the matched suite, FedAvg reaches 0.715/0.809 at $\alpha = 0.1/1$, warm start adds 0.043, and updates cost about 60 MiB. No E3 arm collapses, so the stack is not claimed necessary. Multi-estate, co-observed evaluation remains required.

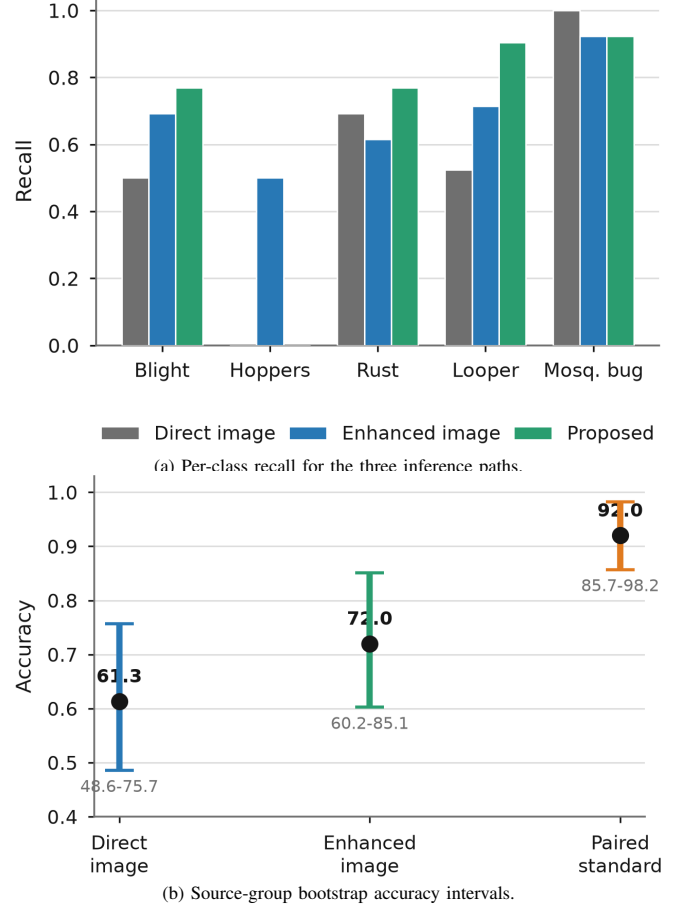


Fig. 13: Class-level gains and uncertainty on the locked 75-crop test. Intervals resample 38 source photographs and do not measure estate shift.

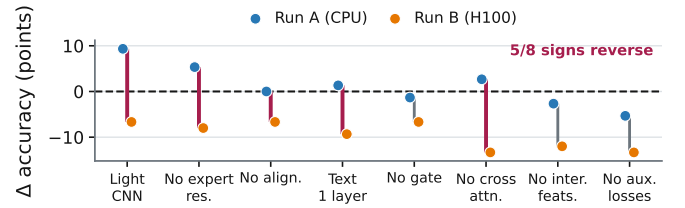


Fig. 14: Paired CPU/H100 component effects. Red connectors cross zero; five of eight signs reverse, so single-run deltas are not ranked.

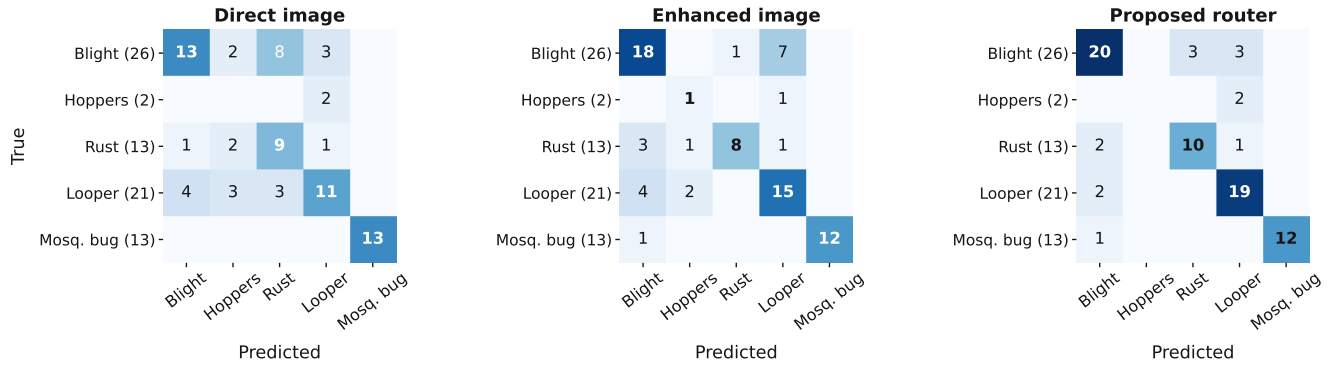


Fig. 12: Where the errors land on the fixed 75-crop test. Each matrix sums to 75; diagonals give the 46, 54, and 61 correct crops of the direct image path, the five-expert enhanced gate, and the exploratory router. All panels share one count colour scale, so shading is directly comparable. The off-diagonal cells are the part no count table records: leaf blight is confused with rust and looper rather than scattered, looper recovers most under the router, and the two leaf-hopper crops fall into looper except for one the enhanced gate recovers.

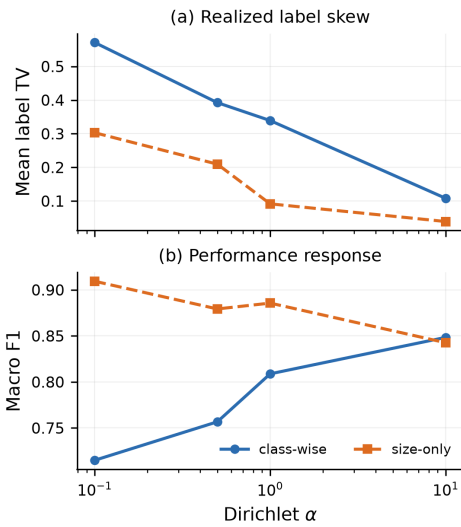


Fig. 15: Partition audit: realized label skew and its macro-F1 response. The legacy size-only splitter stays comparatively IID.

REFERENCES

- J. Yang, G. Xu, M. Yang, and Z. Lin, "Lightweight wavelet-CNN tea leaf disease detection," *PLOS ONE*, vol. 20, no. 5, art. e0323322, 2025.
- Y. Hu, J. Tang, and J. Yang, "Introducing artificial intelligence technology to plant disease management for sustainable agriculture," *Crop Protection*, vol. 184, art. 106764, 2024.
- X. Deng and C. Photong, "Evaluation of tea leaf disease identification based on CNNs VGG16, ResNet50, and DenseNet169," in *Proc. IEECON*, 2024, pp. 1–4.
- K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- D. S. Joseph, P. M. Pawar, and K. Chakradeo, "Real-time plant disease dataset development and detection of plant disease using deep learning," *IEEE Access*, vol. 12, pp. 16310–16333, 2024.
- R. Maurya, S. Mahapatra, and L. Rajput, "A lightweight meta-ensemble approach for plant disease detection suitable for IoT-based environments," *IEEE Access*, vol. 12, pp. 28096–28108, 2024.
- Md. S. H. Shovon *et al.*, "PlantDet: A robust multi-model ensemble method based on deep learning for plant disease detection," *IEEE Access*, vol. 11, pp. 34846–34859, 2023.
- E. A. Al-Shahari *et al.*, "Internet of Things assisted plant disease detection and crop management using deep learning for sustainable agriculture," *IEEE Access*, vol. 13, pp. 3512–3520, 2025.
- W. Shafik, A. Tufail, A. Namoun, M. S. Abu Bakar, and R. A. A. H. M. Apong, "A systematic literature review on plant disease detection," *IEEE Access*, vol. 11, pp. 59174–59203, 2023.
- H. A. Tahir, W. Alayed, and W. U. Hassan, "A federated explainable AI framework for smart agriculture," *IEEE Access*, vol. 13, pp. 97567–97584, 2025.
- H. Devaraj *et al.*, "RuralAI in tomato farming: Integrated sensor system, distributed computing, and hierarchical federated learning," *IEEE Sensors Lett.*, vol. 8, no. 5, pp. 1–4, 2024.
- Q. Ding, X. Yue, Q. Zhang, Z. Xiong, J. Chang, and H. Zheng, "Bc2FL: Double-layer blockchain-driven federated learning for Agricultural IoT," *IEEE Internet Things J.*, vol. 12, no. 4, pp. 4362–4374, 2025.
- P. V. Caminha and H. M. N. da S. Oliveira, "Cloud-driven federated learning for plant disease detection in agriculture," in *Proc. IEEE CloudNet*, 2024, pp. 1–4.
- K. Thonglek and P. Kattanatanrong, "Decentralized federated learning for agricultural plant diseases identification on edge devices," in *Proc. ISAI-NLP*, 2023, pp. 1–6.
- H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, 2017, pp. 1273–1282.
- A. Vaswani *et al.*, "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.
- A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.
- J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL*, 2019, pp. 4171–4186.
- I. Turc, M.-W. Chang, K. Lee, and K. Toutanova, "Well-read students learn better: On the importance of pre-training compact models," *arXiv:1908.08962*, 2019.
- C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. CVPR*, 2016, pp. 2818–2826.
- T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE TPAMI*, vol. 42, no. 2, pp. 318–327, 2020.
- A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.
- J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. ICML*, 2023.
- J.-B. Alayrac *et al.*, "Flamingo: A visual language model for few-shot learning," in *Proc. NeurIPS*, 2022.
- J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "CoCa: Contrastive captioners are image-text foundation models," *TMLR*, 2022.
- J. Lu, C. Clark, R. Zellers, R. Mottaghi, and A. Kembhavi, "Unified-IO: A unified model for vision, language, and multi-modal tasks," in *Proc. ICLR*, 2023.
- Y. Wang *et al.*, "A large language model for multimodal identification of crop diseases and pests," *Scientific Reports*, vol. 15, art. 21959, 2025.
- L. Li, J. Li, D. Chen, L. Pu, H. Yao, and Y. Huang, "FedReplay: A feature replay assisted federated transfer learning framework for smart agriculture," *arXiv:2511.00269*, 2025.
- H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. ICML*, 2021, pp. 10347–10357.
- Z. Liu *et al.*, "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. ICCV*, 2021, pp. 10012–10022.
- Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. CVPR*, 2022, pp. 11976–11986.
- M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.
- V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT," *arXiv:1910.01108*, 2019.
- J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019, pp. 3982–3992.
- P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, 2020, pp. 9459–9474.
- A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, 2019, pp. 8026–8037.
- C. V. Padoiro, T.-W. Chen, T. Komamizu, and I. Ide, "SPQ: Similarity-preserving post-training quantization for portable crop disease detection," *IEEE Trans. AgriFood Electron.*, vol. 4, no. 1, pp. 68–80, 2026, doi: 10.1109/TAFE.2025.3533416.
- S. A. Qadri, N. F. Huang, Y.-H. Huang, and P.-C. Chan, "DDAN-RGAL: A dual-domain adaptive network with reality-gap aware learning for field-deployable plant disease detection," *IEEE Trans. AgriFood Electron.*, early access, pp. 1–18, 2026, doi: 10.1109/TAFE.2026.3667568.
- Q. Tong, N. Zeng, S. Wu, and K. Xu, "Tiny-PMDet: A knowledge-distilled lightweight edge-AI detector for grapevine powdery mildew detection on CMOS sensor based IoT systems," *IEEE Trans. AgriFood Electron.*, vol. 4, no. 1, pp. 383–394, 2026, doi: 10.1109/TAFE.2026.3667753.
- N. Nadeem, M. H. Asad, and A. Bais, "CCUDA-SS: Cross-crop unsupervised domain adaptation for semantic segmentation," *IEEE Trans. AgriFood Electron.*, early access, pp. 1–17, 2026, doi: 10.1109/TAFE.2025.3650340.
- G. M. Dimitri, A. Kocian, F. Scarselli, M. Gori, and S. Chessa, "Continual learning for agrifood: A systematic review," *IEEE Trans. AgriFood Electron.*, vol. 4, no. 1, pp. 117–127, 2026, doi: 10.1109/TAFE.2025.3616733.
- P. Garg, A. Mishra, R. Raja, A. Kumar, M. V. Joshi, and V. S. Palaparthi, "Multimodal data fusion by integrating IoT-enabled sensors and images for Jamun crop disease detection with machine learning," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 582–590, 2025, doi: 10.1109/TAFE.2025.3585065.
- K. K. Keshinemi, S. P. Mohanty, and E. Kougianos, "Semantic-Search: A knowledge-driven classification method for accurate identification of plant diseases," *IEEE Trans. AgriFood Electron.*, early access, pp. 1–11, 2025, doi: 10.1109/TAFE.2025.3619095.
- K. Cao, Y. Yang, L. Xu, X. Wan, L. Yang, and W. Jia, "GAF-Net: An enhanced multidisease for corn leaves detection model based on YOLOv8," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 420–432, 2025, doi: 10.1109/TAFE.2025.3561196.
- S. Das, P. Bikram, A. Biswas, V. C., P. Sinha, and B. B. Bhattacharya, "Transformer-embedded attentive CNN for spectral image analysis of rice blast syndromes," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 615–622, 2025, doi: 10.1109/TAFE.2025.3601808.
- M. A. R. Baee, L. Kane, L. Simpson, Y. Jiang, W. Armstrong, and P. Gauravaram, "Trustworthy digital agriculture: Challenges, countermeasures, and opportunities," *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 518–535, 2025, doi: 10.1109/TAFE.2025.3576623.
- B. Ji *et al.*, "Lightweight tomato leaf intelligent disease detection model based on adaptive kernel convolution and feature fusion," *IEEE Trans. AgriFood Electron.*, vol. 2, no. 2, pp. 563–575, 2024.
- P. S. Thakur *et al.*, "Real-time plant disease identification: Fusion of vision transformer and conditional convolutional network with C3GAN-based data augmentation," *IEEE Trans. AgriFood Electron.*, vol. 2, no. 2, pp. 576–586, 2024.